

Morning Cheat Sheet — read this and nothing else

Java Backend · Infosys · 5 September, 4:00 PM · Learn nothing new today — revision only.

45 minutes. Nothing else. No new material.

If a line here does not ring a bell, open that file for three minutes and come back.

The 8 answers that carry this interview

1. **HashMap**: "`hashCode()` , then a spread of $h \wedge (h \ggg 16)$, then bucket index $(n-1) \& hash$. Collisions chain in a linked list, and since Java 8 a bucket over 8 entries becomes a red-black tree. Default capacity 16, load factor 0.75, so it resizes and rehashes at 12."
2. **equals/hashCode**: "Equal objects must have equal hash codes. Override `equals` without `hashCode` and the map looks in the wrong bucket and never finds the entry."
3. **N+1**: "One query for the list, then one per row for the relation. Fix with JOIN FETCH, `@EntityGraph` , batch fetching, or a DTO projection. That is what took my report from 15 seconds to 6."
4. **@Transactional internal call**: "It works through a proxy. An internal `this.method()` call bypasses the proxy, so no transaction starts."
5. **Constructor injection**: "Fields can be final, dependencies cannot be missing, and I can test with `new Service(mock)` without a Spring context."
6. **Kafka vs RabbitMQ**: "RabbitMQ is a postbox — delivered to a worker, then gone. Kafka is a ledger — events are retained and many consumer groups read independently. I used RabbitMQ for email tasks and Kafka for state-change events with multiple consumers."
7. **Circuit breaker**: "Closed, open, half-open. Failing fast stops the caller exhausting its threads and gives the failing service room to recover."
8. **Index cost**: "Faster reads, but every insert and update maintains every index, so I add them deliberately, not by default."

Core Java

- **Overloading** = compile time, different parameters. **Overriding** = runtime, same signature, subclass. You cannot override a static method — it is hidden.
- **Abstract class** = shared state and code, one only. **Interface** = capability contract, many.
- **ArrayList** = array, O(1) get. **LinkedList** = nodes, O(n) get. Default to ArrayList.
- **HashMap** not thread safe · **Hashtable** locks the whole map, legacy · **ConcurrentHashMap** locks per bucket.
- **String immutable** because of the string pool, security, thread safety, cached hash.

- `==` compares references, `.equals()` compares values. `new String("a") != "a"` .
- **Checked** = compiler forces handling. **Unchecked** extends `RuntimeException` .
- `finally` always runs except on `System.exit()` . Never `return` from `finally` .
- **Fail-fast** throws `ConcurrentModificationException` ; remove via `iterator.remove()` .

Java 8

- **Intermediate** ops are lazy (`filter` , `map` , `sorted`); **terminal** ops trigger (`collect` , `reduce` , `forEach` , `findFirst`).
- `map` = one to one. `flatMap` = one to many, then flattened.
- `Predicate` → filter · `Function` → map · `Consumer` → for each · `Supplier` → lazy value.
- `groupingBy(X, counting())` and `groupingBy(X, summingDouble(Y))` .
- **Optional**: use `map` / `orElse` / `orElseThrow` , not `isPresent` + `get` . `orElse` always evaluates; `orElseGet` is lazy.

Spring Boot

- **IoC** = the framework controls creation. **DI** = how it hands dependencies in.
- Beans are **singleton by default**, so keep them **stateless** — a mutable instance field is a concurrency bug.
- `@Repository` also **translates database exceptions** into Spring's hierarchy.
- `@RestControllerAdvice` + `@ExceptionHandler` = one consistent error shape. Log the stack trace, return a generic message.
- **Auto-configuration** works through `@ConditionalOnClass` and `@ConditionalOnMissingBean` — your own bean always wins.
- `@WebMvcTest` = slice, fast · `@SpringBootTest` = full context · `@MockBean` = replace a bean.

JPA and Hibernate

- **JPA** = spec · **Hibernate** = implementation · **Spring Data JPA** = the layer on top.
- Defaults: `@ManyToOne` and `@OneToOne` are **EAGER**; `@OneToMany` and `@ManyToMany` are **LAZY**. Set `ManyToOne` to **LAZY** explicitly.
- `@Transactional` rolls back on **unchecked** exceptions only, unless `rollbackFor` .
- **Dirty checking** means a managed entity needs no `save()` .
- First-level cache is always on, per transaction. Second-level is opt-in and shared.
- Use `@Enumerated(EnumType.STRING)` , never the **ORDINAL** default.

- Money is **BigDecimal**, never `double`.

Microservices

- Gains: independent deploy, independent scale, fault isolation. **Costs: network failure, no distributed transaction, harder debugging.** Say the costs.
- **Eureka** = discovery · **API Gateway** = single entry, auth and routing in one place · **Config Server** = externalised config.
- **Saga** = local transactions plus **compensating** transactions. Eventual consistency.
- Only retry **idempotent** operations. Always set a **timeout**.
- Split by **business capability**, not technical layer.
- Observability: centralised logs, **correlation ID** tracing, metrics.

Messaging

- Kafka: **topic** → **partitions** → **offsets**; order is guaranteed **within a partition**, so key by transaction or account ID.
- **Same consumer group** = share the partitions. **Different groups** = each gets everything.
- **At-least-once** is the normal default, so consumers must be **idempotent**.
- RabbitMQ: producer → **exchange** → queue. Direct, topic, fanout. **DLQ** after N failed attempts.

Security

- **JWT** = header.payload.signature. **Signed, not encrypted** — anyone can read the payload, so put nothing sensitive in it.
- Stateless, so any instance validates it — that is why it suits microservices.
- Cannot easily revoke: short-lived access token plus refresh token, or a blocklist.
- CSRF is off for a stateless API because the browser does not auto-attach an `Authorization` header.
- **BCrypt** for passwords — slow and salted by design.
- `hasRole("ADMIN")` looks for the authority `ROLE_ADMIN`.

SQL

- **Left-most prefix rule:** an index on `(status, created_at)` does not serve `created_at` alone.
- Index killers: a **function on the column**, and a **leading %** in LIKE.
- **EXPLAIN**: `type: ALL` means full scan · check `key`, `rows`, and `Extra`.
- **WHERE** filters rows **before** grouping; **HAVING** filters groups **after**.
- **UNION** deduplicates and sorts; **UNION ALL** does not and is faster.
- Customers with no transactions = `LEFT JOIN ... WHERE t.id IS NULL`.
- Deep **OFFSET** is slow — prefer **keyset pagination**.

Coding round

- Talk while you type. Ask about edge cases **first**. State the complexity at the end.
- `LinkedHashMap` for first-non-repeating. `seen.add(n)` returns false for a duplicate.
- Two sum = HashMap, O(n), trading space for time. Mention the brute force first.
- `Integer` cache is -128 to 127, so `128 == 128` is **false** for boxed Integers.

Your stories — the numbers

15s → 6s report: N+1 found via SQL logging, fixed with a projection query, indexes, rewritten JOINS. *"My first guess was wrong, which is why I measure first."*

2-3s → under 1s APIs: N+1 plus lazy loading plus indexing.

RabbitMQ: email off the request thread, bulk approvals stopped timing out.

Kafka: state changes to multiple consumer groups, retained and replayable.

RBAC: requester, approver, admin — enforced server side, because a UI check only hides the button.

Classifieds: 3 services, own databases, Eureka, API Gateway, JWT, global exception handling.

The three questions you must not fumble

"Tell me about yourself" → Now → What you built → Why this role. Under 2 minutes.

"Why did you move from recruitment to development?" → B.Tech in CS, JSPIDERS training, the recruitment year was technical recruitment, moved to engineering Dec 2024.

Then: *"That year made me comfortable in client and requirement conversations, which is a lot of what this role is."*

"How much experience do you have?" → **Answer with dates, never a rounded number.**

"Engineering since December 2024, so about a year and nine months, plus a year in technical recruitment before that."

Questions to ask them "Which domain would this role sit in?" · "Greenfield microservices or modernising a monolith?" · "What would a successful first three months look like?" ·

"What is the biggest technical problem this hire would help with?"

The last five minutes

- You have real production numbers and banking domain experience. Most candidates for this role have neither.
- **If you do not know something: say so, then reason out loud.** That has never cost anyone an offer. Bluffing has.
- Give the **why**, not just the **what**. "HashMap stores key-value pairs" is a textbook answer; "O(1) because it hashes to a bucket instead of scanning" is an engineer's.

- Sit up. Breathe out slowly. Smile before you speak — it changes your voice.

Go get it.